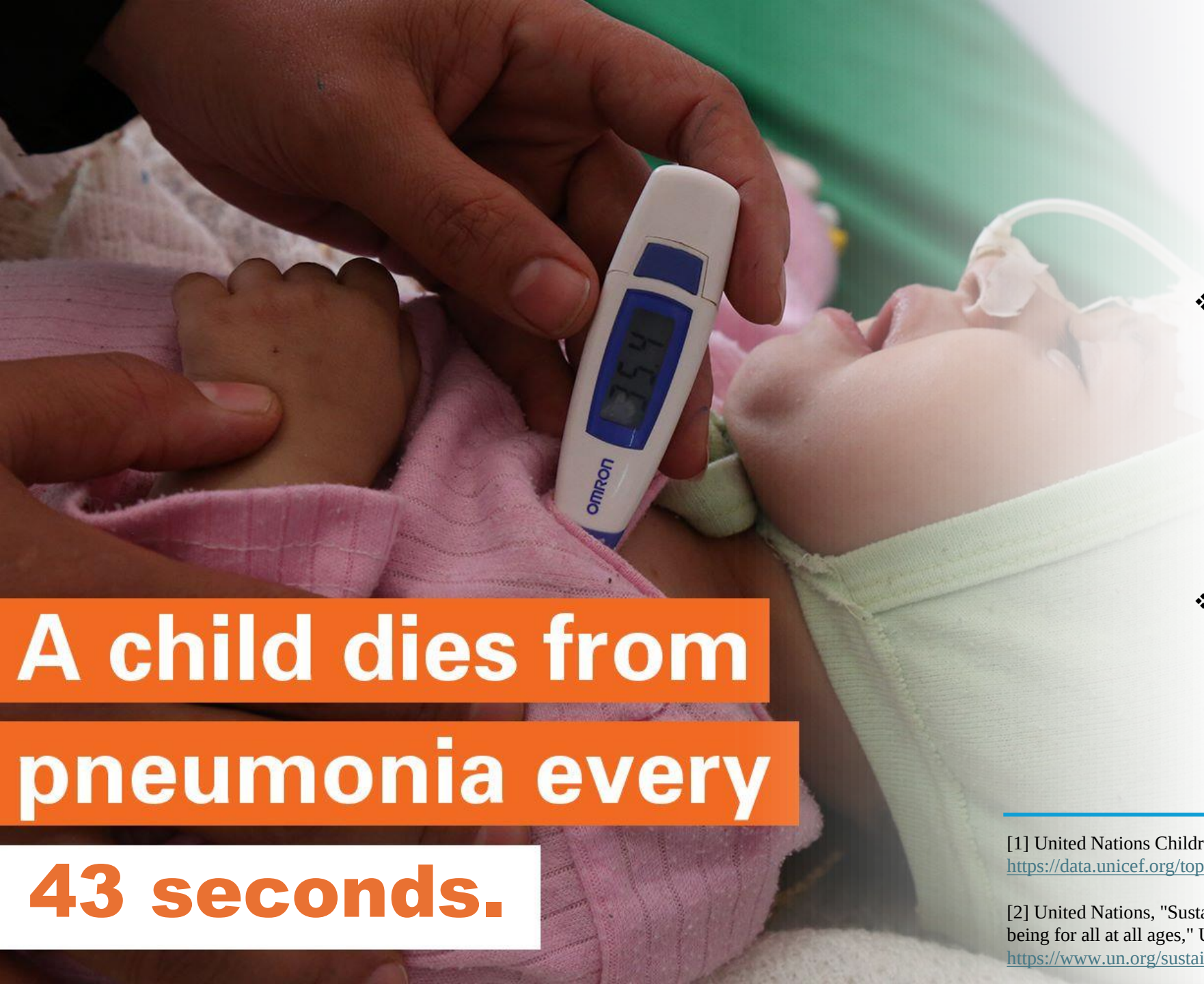


AI-Powered Pneumonia Detection in Low-Resource Settings



- ❖ **Student:** Evans Konadu
 - ❖ **Supervisor:** Professor Dimitrios Makris
 - ❖ **Date:** September 24, 2025
-



**A child dies from
pneumonia every
43 seconds.**

Background & Motivation

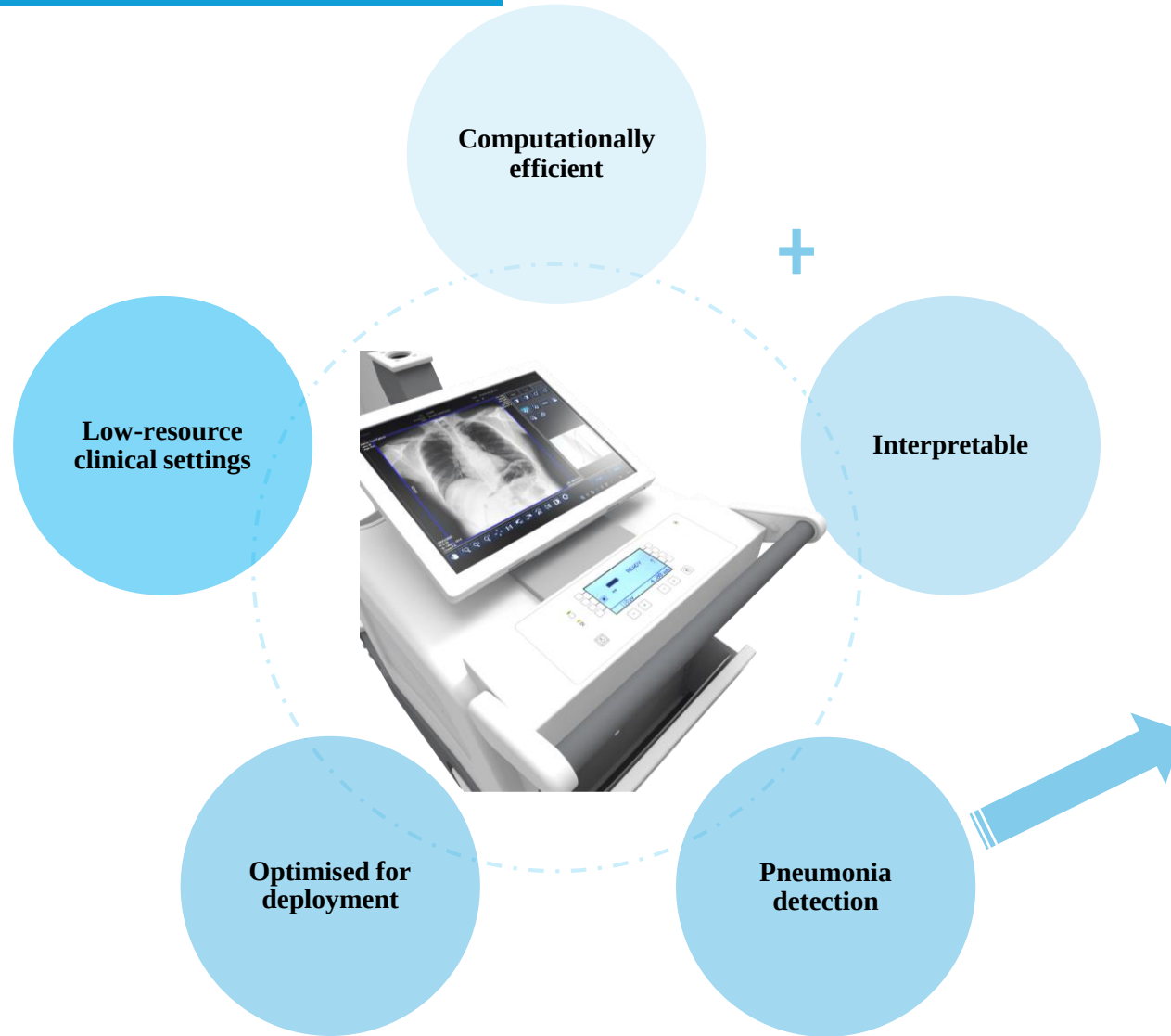
- ❖ **Background:** Pneumonia kills more children than any other infectious disease, claiming the lives of over **700,000** children under 5 every year, or around **2,000** every day, with limited radiologist access in rural clinics hindering diagnosis[1].
- ❖ **Motivation:** To meet SDG 3.2.1 targets for reducing child mortality by 2030, effective pneumonia detection tools are critical, yet existing deep learning models require substantial computational resources, making them impractical for low-resource settings[2].

[1] United Nations Children's Fund (UNICEF), "Pneumonia," [Online]. Available: <https://data.unicef.org/topic/child-health/pneumonia/>. [Accessed: July 10, 2025].

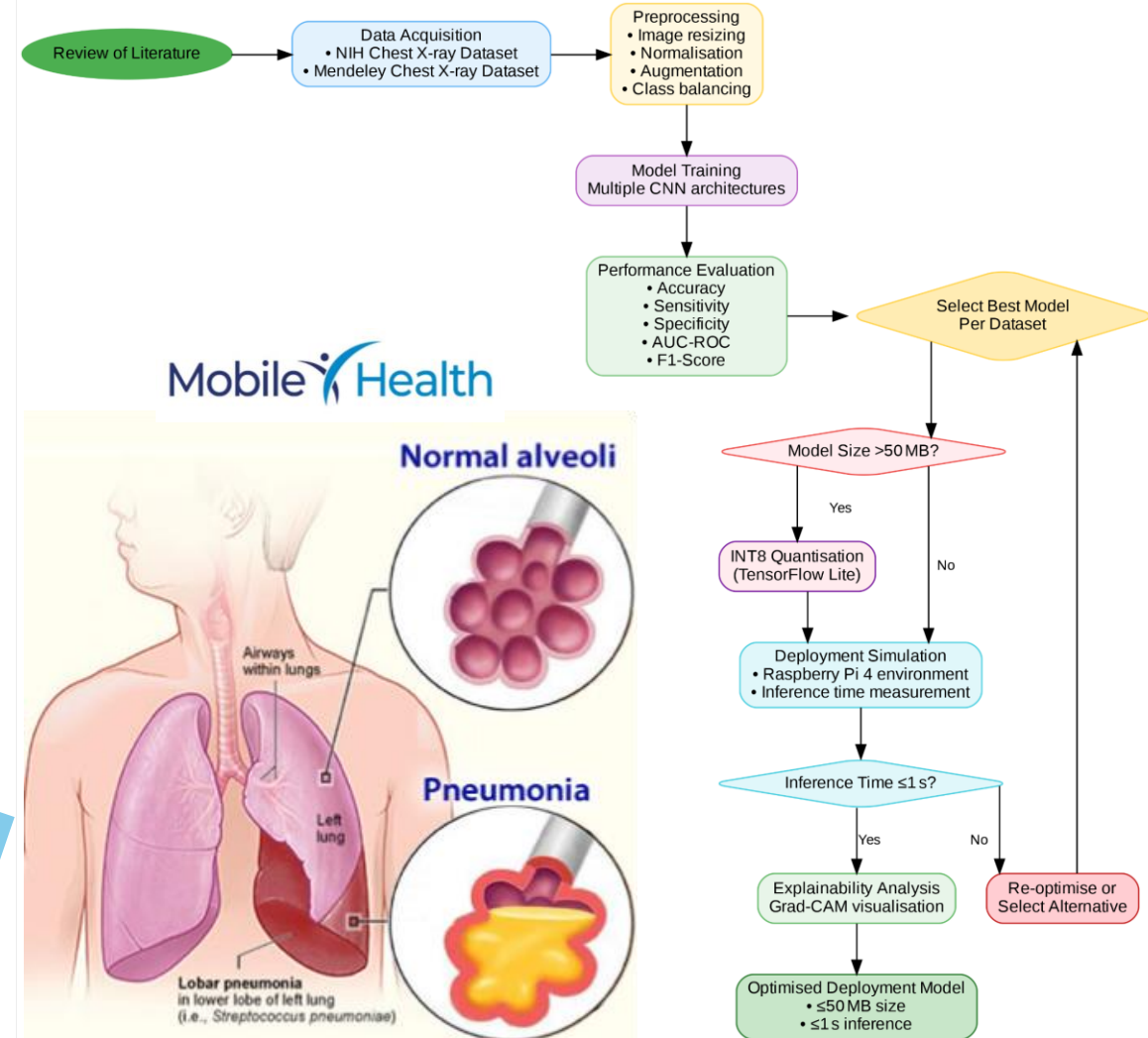
[2] United Nations, "Sustainable Development Goal 3: Ensure healthy lives and promote well-being for all at all ages," UN Sustainable Development, Oct. 2023. [Online]. Available: <https://www.un.org/sustainabledevelopment/health/>

Aim

This project seeks to develop DL models:

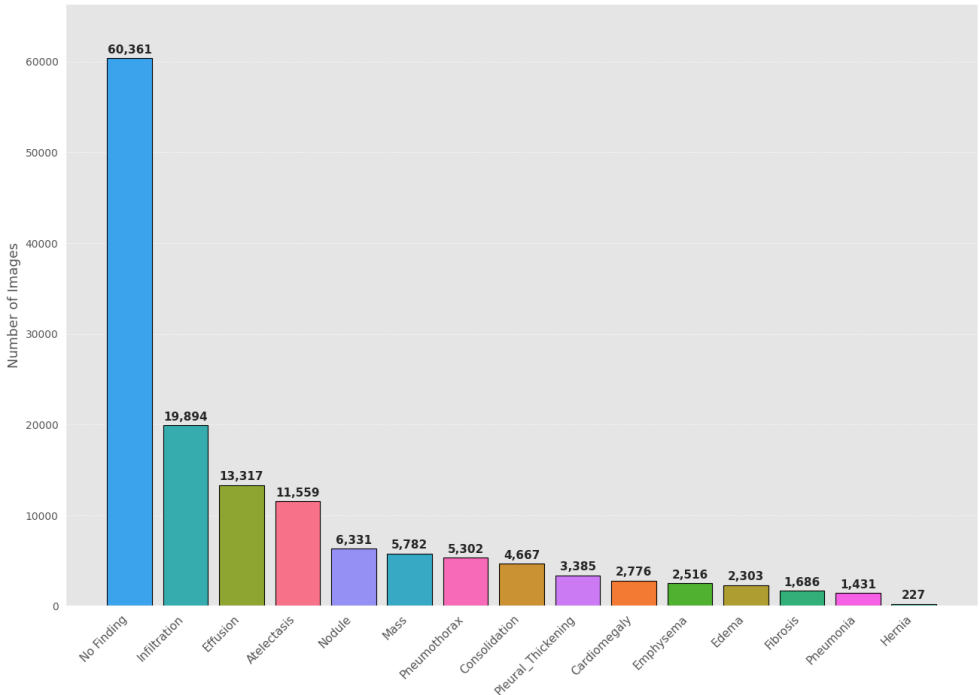


Methodology Flowchart

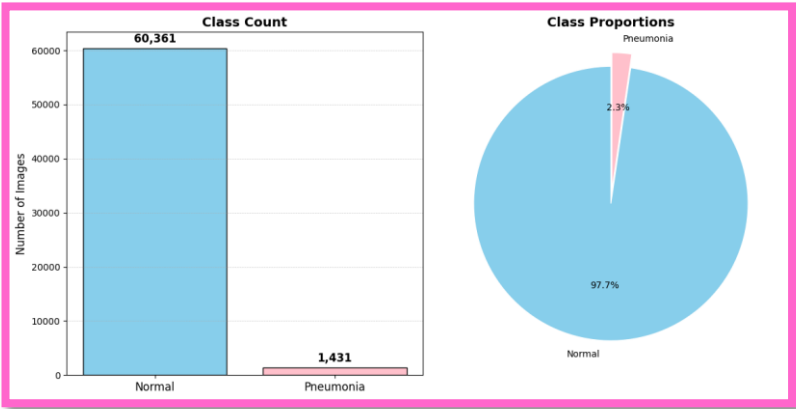
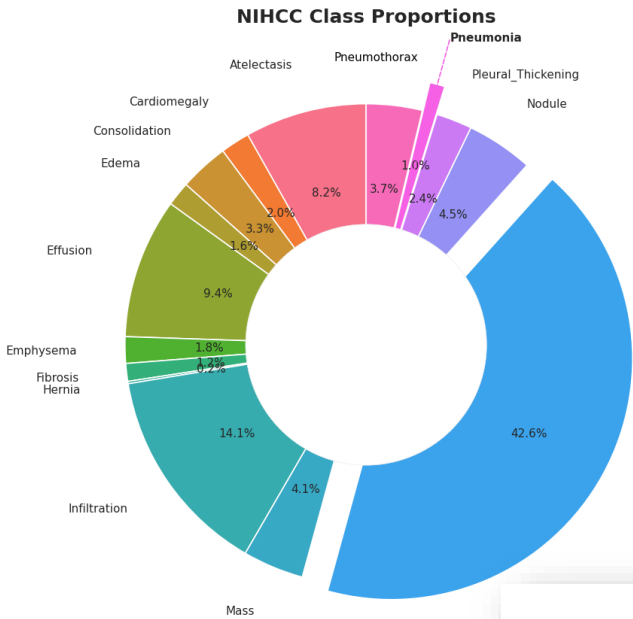


Datasets Overview

❖ NIH ChestX-ray14 Dataset: NIHCC Dataset Distribution



Distribution of 14 Labels in NIH ChestX-ray14 Dataset



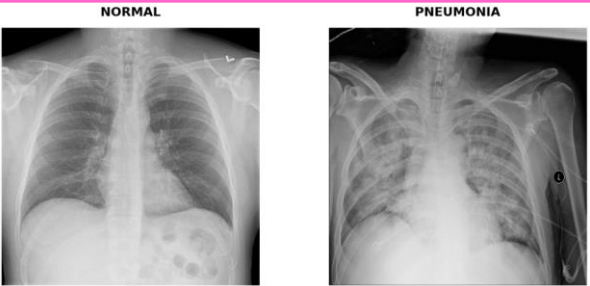
Bar chart and pie chart highlighting the imbalance (60,412 Normal vs. 1,431 Pneumonia: **Ratio 42:1**)

Dataset Summary:

- Total Images: 112,120
- Unique Findings: 15
- Most Common: No Finding (60,361)
- Rarest: Hernia (227) and Pneumonia: (1,431)

Bar chart with annotated counts and pie chart with percentage annotations showing all 14 labels.

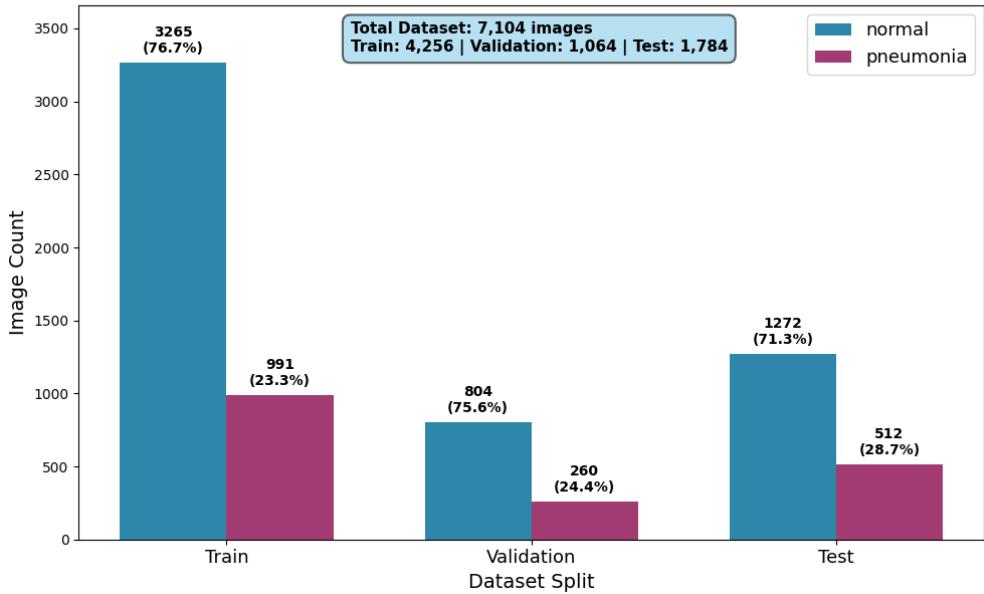
Sample NIH X-Ray Images



NIH Dataset Strategy

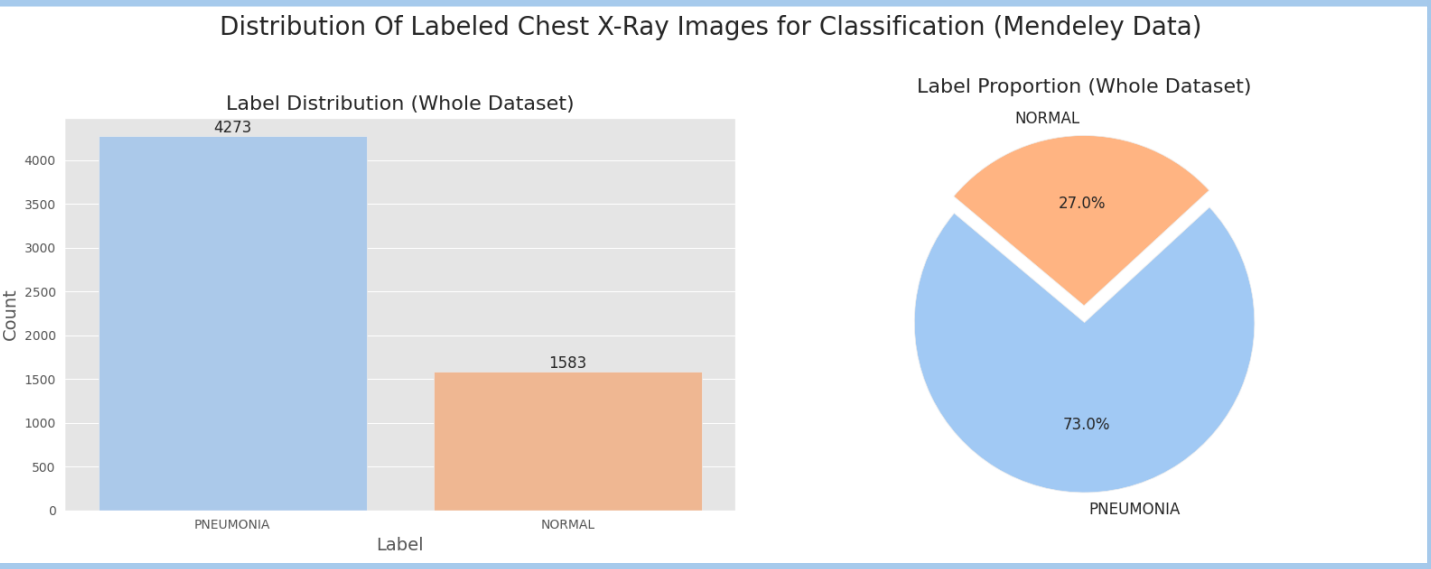
Cluster-based undersampling reduced severe class imbalance from 42:1 to **3:1** ratio

Class Distribution Across Train, Validation, and Test Sets NIH Chest X-Ray Dataset (PNEUMONIA vs NORMAL)

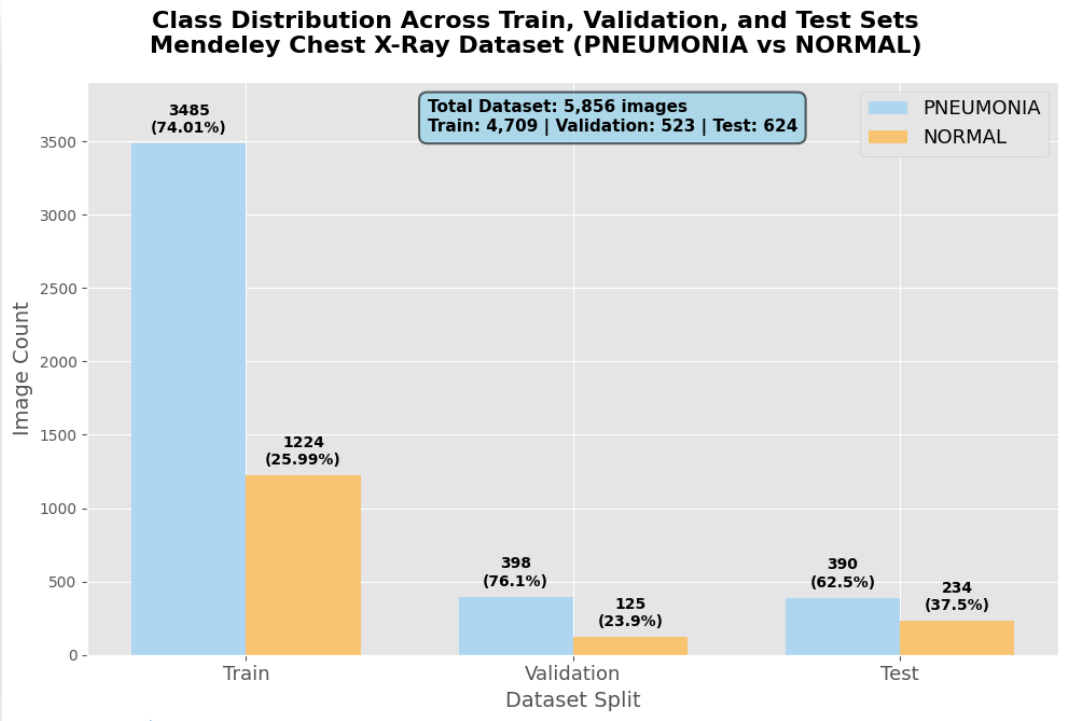


❖ Mendeley Data (Kermany et al.):

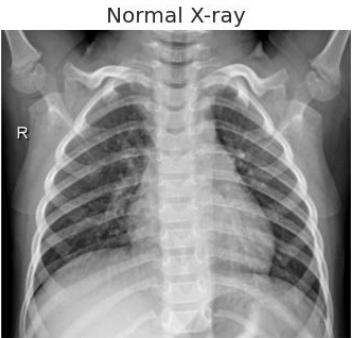
Distribution of Labeled Chest X-Ray Images (Mendeley Data)



Bar chart with counts and pie chart with percentages for Normal and Pneumonia classes.



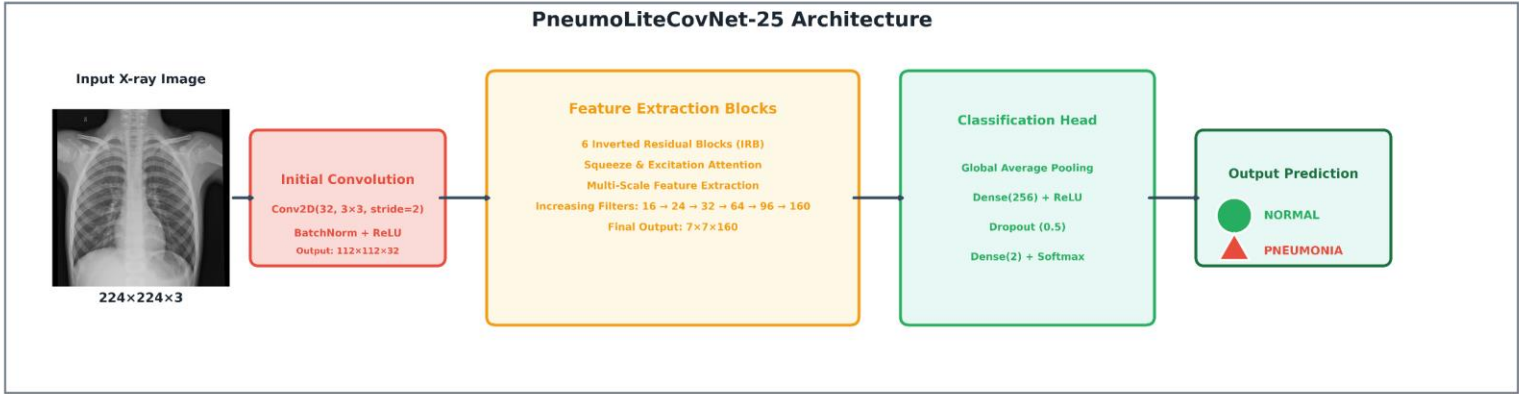
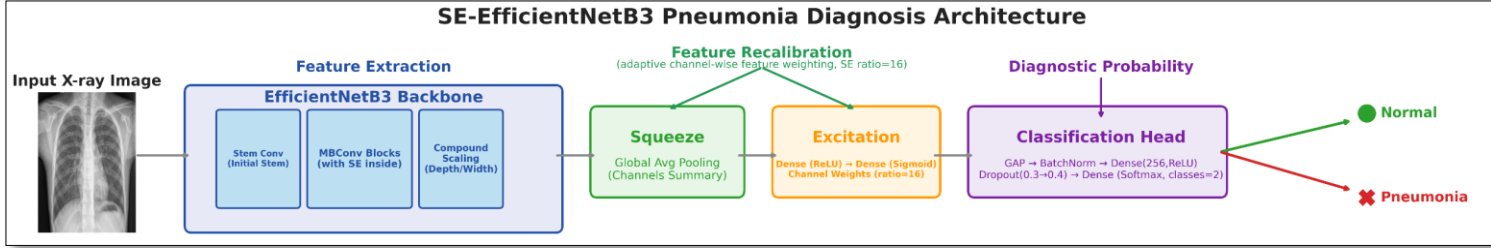
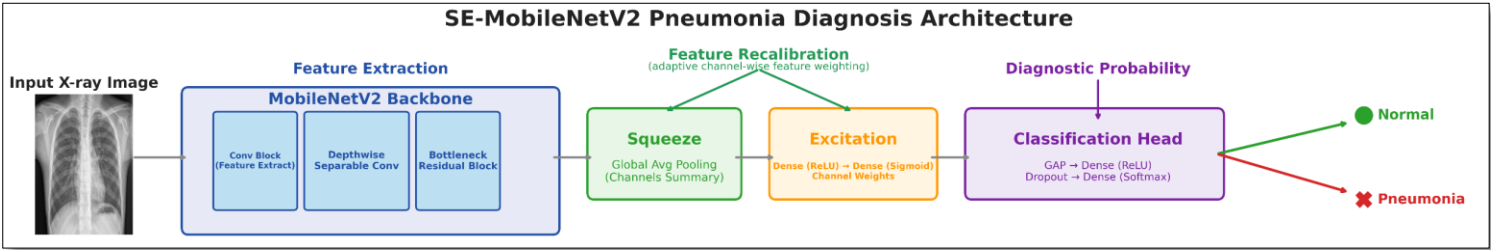
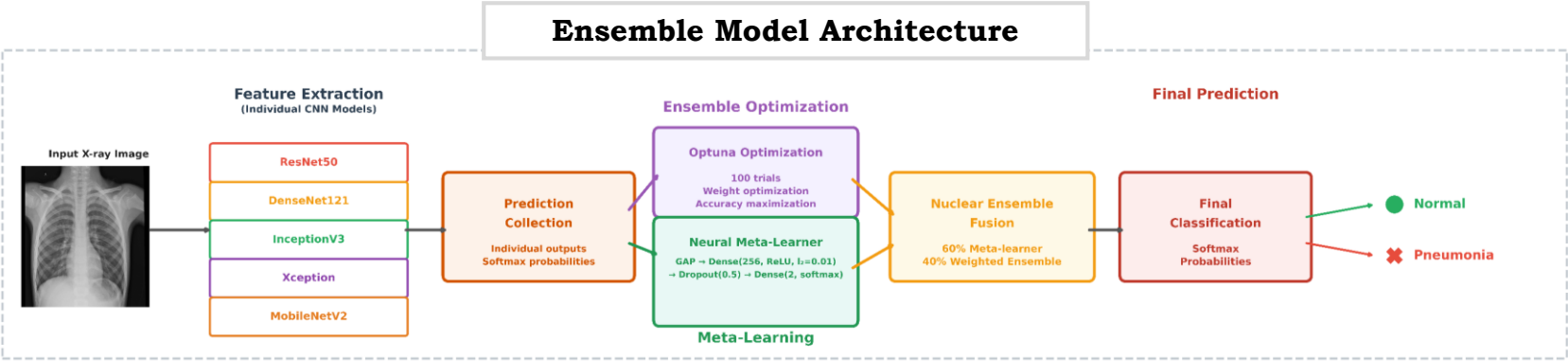
Sample X-Ray Images from Mendeley Dataset



Mendeley Dataset Strategy

Utilising pre-existing train/test structure, 10% internal validation split

Novel Architecture Contributions Visuals



COMPARATIVE PERFORMANCE METRICS OF MODELS
ACROSS DATASETS (Pre-Quantisation Analysis)

(Bold = Best | Underline = 2nd Best)

Dataset	Model	Accuracy (%)	Precision (w.AVG) (%)	Sensitivity (%)	Specificity (%)	F1-Score (w.AVG) (%)	AUROC
Mendeley Data (Paediatric) - Kermany et al. (2018)	Nuclear Ensemble	<u>96.79</u>	98.18	96.67	97.01	<u>97.42</u>	<u>0.9915</u>
	Xception	96.96	<u>96.98</u>	96.92	97.01	97.55	0.9933
	EfficientNetB4	95.19	95.23	97.95	90.60	95.16	0.9887
	SE-MobileNetV2	94.07	94.06	95.90	91.03	94.06	0.9851
	MobileNetV2	91.51	91.69	<u>96.92</u>	82.48	91.38	0.9818
	SE-EfficientNetB3	92.47	92.52	96.41	85.90	92.40	0.9818
	DenseNet121	91.51	91.73	97.18	82.05	91.37	0.9789
	VGG16	89.26	90.71	99.74	71.79	88.81	0.9815
	ResNet50	87.50	89.28	82.82	95.30	87.68	0.9705
	InceptionV3	87.50	88.06	96.41	72.65	87.13	0.9507
	PneumonLiteCovNet-25	82.85	83.37	94.10	64.10	82.19	0.9137
NIH ChestX-ray14 (Adults) - Wang et al. (2017).	Nuclear Ensemble	78.70	78.10	38.67	94.84	76.24	<u>0.8528</u>
	Xception	<u>79.54</u>	<u>78.61</u>	47.27	92.53	<u>78.09</u>	0.8364
	EfficientNetB4	86.10	86.10	78.91	88.99	86.22	0.8986
	SE-MobileNetV2	74.38	75.80	63.67	78.69	74.92	0.7854
	MobileNetV2	74.05	75.70	64.26	77.99	74.66	0.7833
	SE-EfficientNetB3	73.77	71.42	33.59	89.94	71.35	0.7399
	DenseNet121	71.69	73.05	58.20	77.12	72.24	0.7447
	VGG16	75.56	76.07	<u>60.74</u>	81.53	75.79	0.7949
	ResNet50	72.53	72.93	54.49	79.80	72.72	0.7423
	InceptionV3	73.71	72.84	48.24	83.96	73.71	0.7519
	PneumonLiteCovNet-25	65.86	70.88	63.28	66.90	67.31	0.6896
	CheXNet (DenseNet-121)						0.7679

Model Optimisation via Quantisation

❖ Quantisation Approach:

Ensemble/Xception model (Mendeley Dataset)

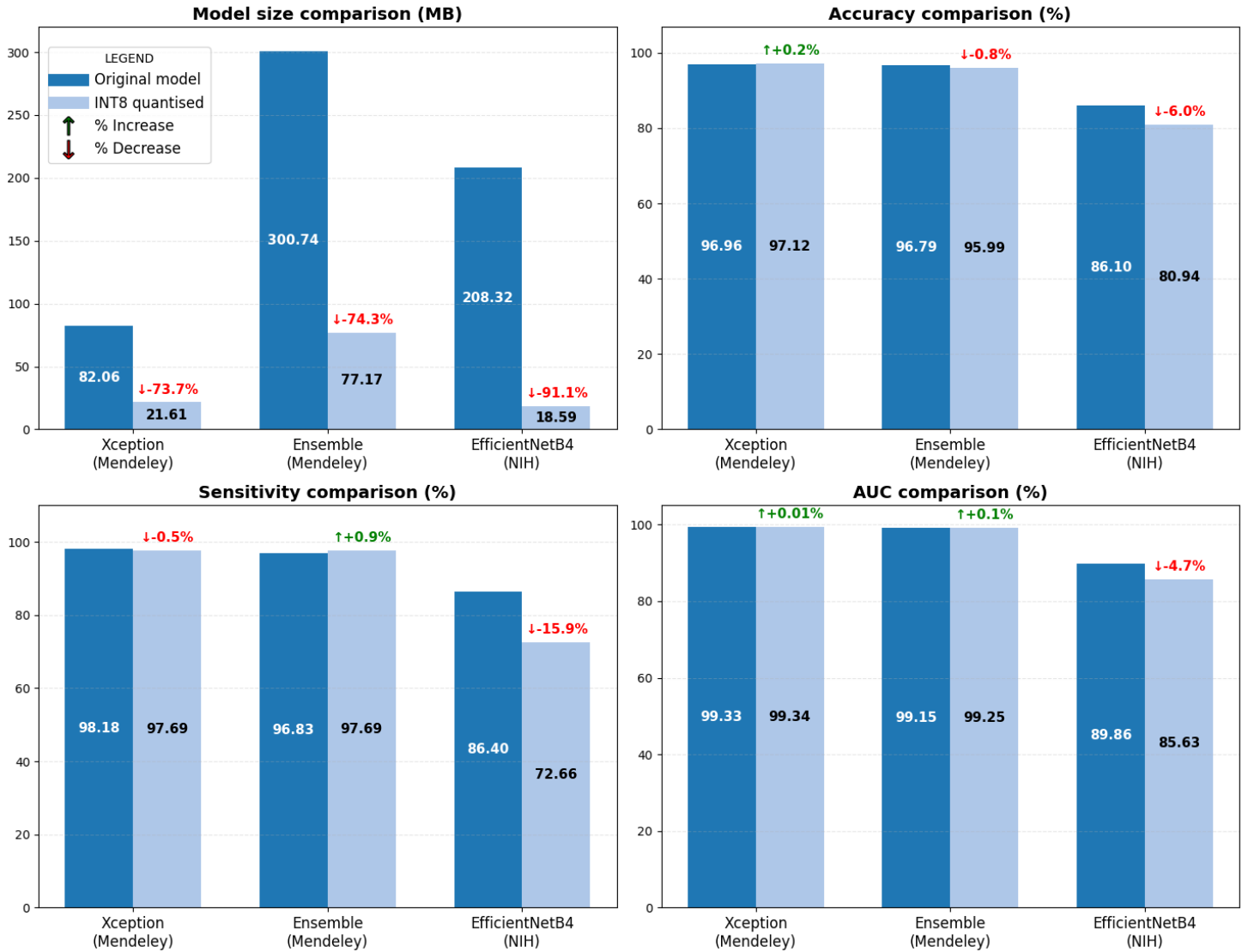
- PT-PQ (INT8 parameter quantisation; FP32 input/output conversion), inspired by S. Thakur *et al.*

EfficientNetB4 Model (NIH Dataset)

- Hybrid (dynamic range) INT8, which quantises only weights to INT8 while keeping activations in FP32.

Model	Size (MB)	Acc %	Sens %	Spec %	Prec (w) %	F1-Score (w) %	AUC
Orig. Xcep.	82.08	96.96	96.92	97.01	98.18	97.55	0.9933
Quant. Xcep.	21.61	97.12	97.69	96.15	97.69	97.69	0.9934
Orig. Ensemble	300.74	96.79	96.67	97.01	96.83	96.80	0.9915
Quant. Ensemble	77.17	95.99	97.69	93.16	95.97	96.82	0.9925
Orig. EffNetB4	208.32	86.10	78.91	88.99	86.40	86.22	0.8986
Quant. EffNetB4	18.59	80.94	72.66	84.28	65.03	68.63	0.8563

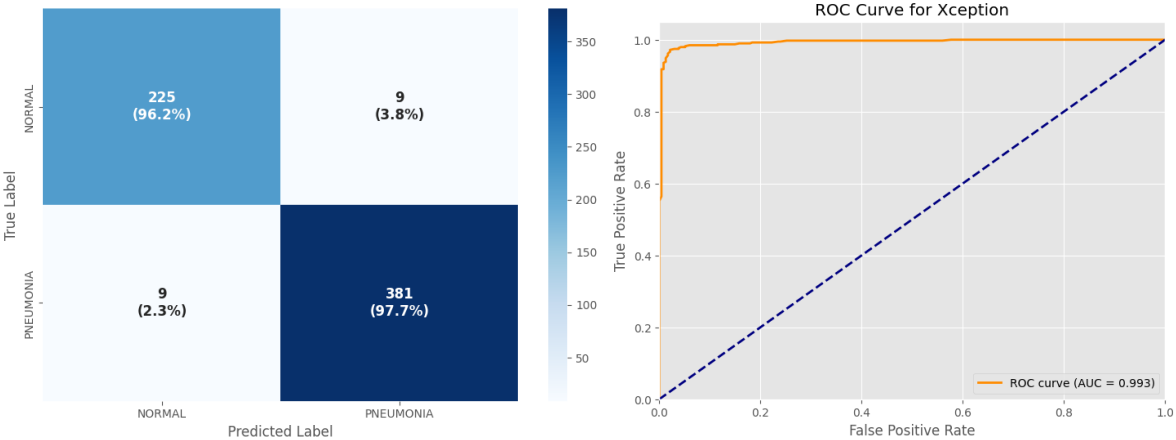
Model Performance: Original vs INT8 Quantised



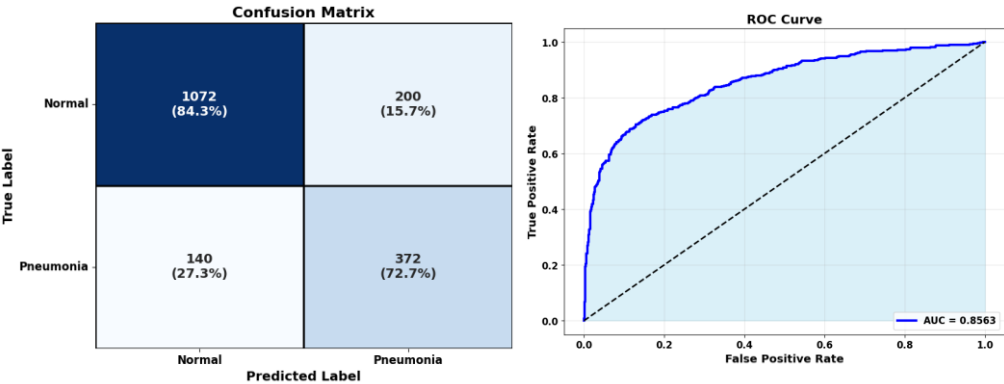
Deployment on Edge Devices

Metric	Colab CPU (Windows Laptop)	Raspberry Pi 4 Model B (4 GB)
CPU SoC	Intel x86_64 CPU @ 2.2 GHz	Broadcom BCM2711 SoC (Cortex-A72 Quad-Core ARM v8) @ 1.5 GHz
Available Cores	1 physical core, 2 logical thread	4 physical cores, 4 logical threads
RAM & Arch.	12.67 GB LPDDR4, x86_64	4 GB LPDDR4-3200, ARM 64-bit

Confusion Matrix And AUC Curve – Xception (Mendeley Dataset)



Confusion Matrix And AUC Curve – EfficientNetB4 (NIH Dataset)



Edge Deployment Performance: Size vs Inference

Model	Dataset	Size (MB)		Adj. Inference Time (ms)	
		Orig.	Quant.	Orig.	Quant.
Ensemble	Mendeley	300.74	77.17	5230.78 ± 828.11	1646.77 ± 409.74
Xception	Mendeley	82.06	21.61	1038.79 ± 205.36	550.76 ± 119.67
EffNetB4	NIH	208.32	18.59	1461.07 ± 425.52	232.69 ± 93.50

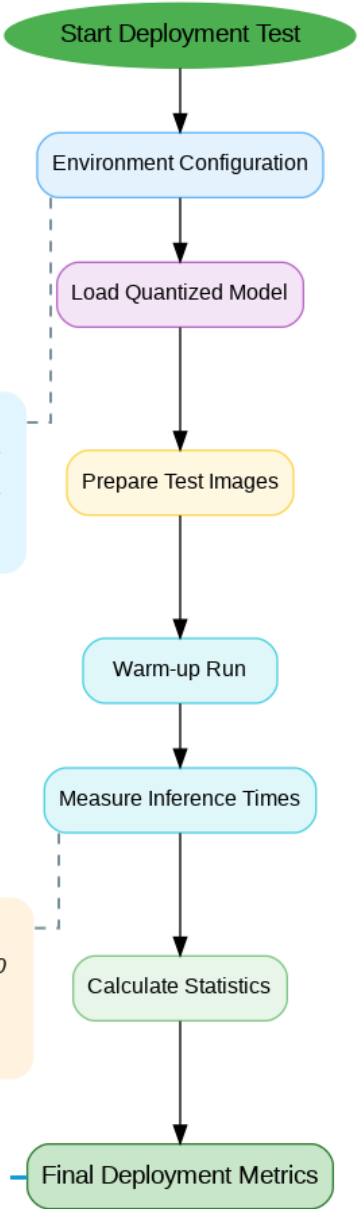
CPU Thread Configuration

Set intra-op threads: `os.environ["TF_NUM_INTRAOP_THREADS"] = "4"`
Set inter-op threads: `os.environ["TF_NUM_INTEROP_THREADS"] = "1"`
Mimics Raspberry Pi 4-core CPU

Clock Speed Adjustment

- Get current CPU frequency `current_ghz = psutil.cpu_freq().current / 1000`
- Calculate adjustment factor `clock_factor = current_ghz / 1.5`
- Apply time adjustment `adjusted_time = raw_time × clock_factor`

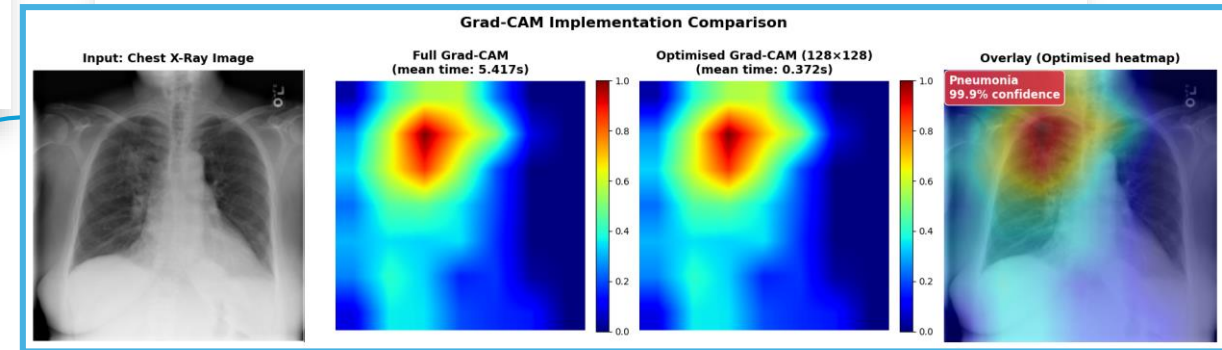
Edge Deployment Workflow



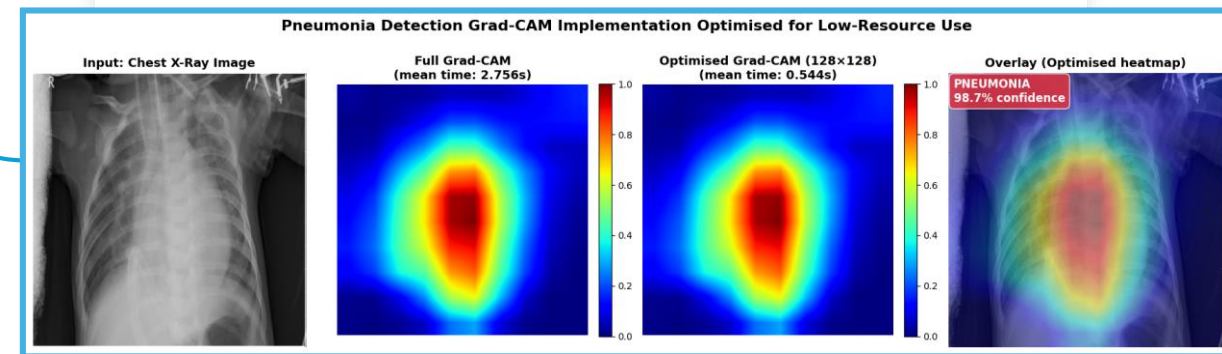
Explainable AI: Grad-CAM Visualisations for Model Interpretability

- Optimised Grad-CAM implementation achieving 5-15× speedup for real-time clinical deployment
- Efficient gradient computation using 128×128 downsampling with bilinear upsampling for visualisation
- Custom implementation enables sub-second explanation generation on edge devices
- Red/yellow regions highlight pneumonia-relevant features (regions of greatest concern) for clinical interpretation
- Real-time explanations enhance physician trust and support clinical decision-making

EfficientNetB4 Model – NIH Chest X-ray Dataset



Xception Model – Mendeley Dataset



❑ Challenges & Solutions

- 1. Optimising model without losing performance
 - Implemented INT8 PT-PQ and INT8 Hybrid quantisation.
- 2. Testing real-world deployment without physical hardware
 - Reconfigured Colab CPU to mimic Raspberry Pi 4 (1.5GHz clock, 4 threads)

❖ Key Outcomes:

- Xception optimal for paediatric screening (97.12% accuracy, 21.61 MB)
- EfficientNetB4 optimal for adult populations (80.94% accuracy, 18.59 MB)
- 74-91% model size reduction with maintained clinical performance

❑ Deployment Readiness

- Edge-optimised models ready for resource-constrained environments
- Real-time inference with integrated explanations

❑ Future Directions

- Integration with mobile diagnostic platforms for field deployment
- Hardware-aware optimisation for diverse edge computing environments



• Q & A



**THANK
YOU**